

From Concepts to Measures

How Do We Measure the Concepts We Define?

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Where We Left Off

- **Datum:** singular; one individual piece of raw information.
- **Data:** plural; a collection of pieces of raw information along a characteristic.
- **Dataset:** a collection of data on multiple characteristics.
- **Database:** a collection of datasets.

A statistic is calculated and interpreted using multiple data points:

1. Can you see uncertainty? (i.e., Question averages)
2. Can you see yourself in the data? (i.e., Get context)
3. How was the data collected? (i.e., Check the questions)
4. Look at the data yourself!

What Are You Even Talking About?

Concepts are social constructs made explicit and specific for the purpose of research.

Conceptualization is then the process through which concepts are made more precise.

The **conceptual definition** is the clear description of a concept that helps identify its measurable properties the unit of analysis to which the concept applies.

What's a unit of analysis?

The basic building block of what the study is studying.

- The entity (person, city, county, university, state, school district, school, year, country, etc.) we want to analyze.
- The thing to which the concept applies (the who/what of the study).

Can be simple (e.g., country) or complex (e.g., country-conflict-party-year).

Good Concepts Are...

Observable: measurable with our ordinary senses.

- Directly observable: place of residence; weight; age.
- Indirectly observable: social status; regime type.
- Unobservable: human nature; divine intervention.

Precise: clear about the properties it summarizes; refers to only **one** set of properties of a given phenomenon.

- E.g.: “*level* of education” vs. “education.”
- What are other dimensions of education?

Develop conceptual definitions for two of these things:

1. poverty
2. seriousness of crime
3. social status
4. academic achievement
5. conservative

Your definition should **specify the unit of analysis** and **clarify the properties** of the phenomenon encapsulated in the concept

Operationalization Is Hard, Too

Operationalization: the process of turning concepts into *operational* measures. The process of turning “squishy” concepts into “firm,” consistent, comprehensible, measurable observations.

Sure, our conceptual definition of democracy has key components to it, but how do we *measure* those components in practice?